

Continuous liquid-phase valorization of bio-ethanol towards bio-butanol over metal modified alumina



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ABSTRACT

Commercial mixed-phase aluminum oxide was used as a heterogeneous catalyst support, providing slightly basic properties which are well-suited for the condensation of bio-ethanol to C₄ hydrocarbons, such as 1-butanol. Different metals (Cu, Ni and Co), at various metal loadings were deposited on the support. Consequently, the catalytic reactions were carried out in a continuous laboratory-scale fixed bed reactor operated at 240 °C and 70 bar. The catalysts were characterized by means of XRD, TEM, FT-IR, XPS and ICP-OES. Different metals were found to give entirely different product distributions. With the best catalysts, the selectivities towards 1-butanol close to 70% were reached, while the ethanol conversion typically varied between 10 and 30% – strongly depending on the metal applied. It was observed that low loading of copper and high loading of nickel were responsible for the formation of 1-butanol, whereas cobalt and high loading of copper resulted in the production of ethyl acetate. The reaction was found to be extremely sensitive to catalyst preparation conditions and procedures such as metal loading, calcination/reduction temperature and, thereby, to the formation of corresponding crystallite structure.

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1. Introduction

Bio-ethanol is one of the most widely discussed options for replacing traditional fossil fuels. It is well known that countries in the tropical zone, e.g. Brazil have implemented a high-level sugarcane-based ethanol production technology by applying farmland. In the Northern hemisphere, Nordic countries have remarkable forest resources which can be used for manufacturing of ethanol in pulp mills by taking advantage of the applicable polysaccharides in wood. Forest-based (lignocellulose) residual approach is well-known for its ability to serve as a non-edible source of sugars, not competing with human-related food production. Despite the high

interest in bio-ethanol, there are certain key issues that should be addressed.

Less than perfect physico-chemical properties of ethanol restrict its use as a fuel additive in conventional engines designed for fossil fuels, at least in higher portions (exceeding 10%) [1,2]. Some of the problematic features of ethanol are worth mentioning, such as its relatively low energy content (in comparison to e.g. gasoline), high water solubility and corrosive behavior. These arguments support the idea of developing a more suitable candidate for replacing ethanol. A plausible approach is based on the condensation of ethanol leading mainly to its four-carbon atom counterpart, 1-butanol. 1-Butanol resembles closely traditional gasoline in terms of its physico-chemical properties [1,2] and is, thus, suitable in much higher blending levels with gasoline [3].

Historically, two production processes for 1-butanol have dominated, a process either based on fossil raw materials (oxo- and aldol process) [4] or a bio-based method [5], such as the Acetone-Butanol-Ethanol (ABE)-fermentation. Today, there are numerous ongoing research efforts that deal with different bio-related pathways leading to the formation of 1-butanol [4–6]. 1-Butanol is not only valuable as a fuel component but it can also be utilized in the

Abbreviations: ABE, acetone butanol ethanol; GC-FID, gas chromatography-flame ionization detector; HPLC, high pressure liquid chromatography; GC, gas chromatography; TEM, transmission electron microscopy; XRD, X-ray diffraction; FT-IR, Fourier transform infrared spectroscopy; XPS, X-ray photoelectron spectroscopy; ICP-OES, inductively coupled plasma-mass spectrometer; UV–Vis, ultra violet–visible diffusion reflectance spectroscopy.

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manufacture of different esters, ethers, acetates and resins [5]. Additionally, other ethanol derivatives (e.g. ethyl acetate, 1,1-diethoxy ethane) have shown some potential as fuels for diesel engines [7–9].

From a viewpoint of modern chemical engineering, the straight one-pot continuous valorization of bio-ethanol to bio-butanol would be the most feasible route. There are a few scientific reports available presenting the desired reaction pathway [10–26]. The use of mixed alumina based oxides [10–16], γ -alumina [17], hydroxyapatite structures [18–21], magnesia [22,23], alkali cation zeolites [24], commercial cobalt powder [25] and sodium modified zirconia [26] have been demonstrated. Based on the structures of these materials, some conclusions can be drawn. For instance, low valence state metal cations form basic metal oxide structures seem to promote the reaction. Corresponding higher valence state oxides are known to have acidic characteristics. Also crystallite molecular structures seem to be a prerequisite. Additionally, the evaluation of the reaction mechanism has two different approaches. Some previous studies [17,22,24] suggest that ethanol condensation proceeds via a straight dehydration pathway, whereas a newer approach [10–16,18,21,23,26] supports the idea of aldol condensation. However, the temperature of the operating conditions may also have an impact on the reaction mechanism [20]. It is also important to evaluate the dominating by-product, ethyl acetate, that can be manufactured over similar kind of catalysts as can be used in the butanol synthesis [27–31] but also on e.g. silica catalyst [32]. We have found that especially cobalt supported on alumina indicates a tendency to form ethyl acetate.

In our previous publication [33] covering qualitative screening of alumina supported catalysts, more evidence was obtained which supports the condensation type reaction mechanism and, at the same time, the significance of by-products essential for the reaction were identified. Hereby we demonstrate a continuous liquid phase one-pot synthesis strategy based on an extensive alumina based catalyst optimization. A more profound view on the engineering aspects of bio-based ethanol valorization is available [34,35].

2. Materials and methods

2.1. Catalyst preparation

All alumina supported metal catalysts were prepared with the deposition-precipitation method based on the hydrolysis of urea, aiming at producing small and uniform metal cluster size distributions. In a typical procedure, a commercial mixed phase (α , β , γ , θ) alumina support (Euro Support, A 201) was crushed and sieved to 150–250 mesh fraction. The iso-electric point of the support is known to be approximately pH = 8.4, which indicates a slightly basic behavior. Thereafter, weighted amounts of metal nitrate (NiNO_3 , CoNO_3 , CuNO_3), urea and alumina were mixed in deionized water under magnetic stirring. The solution was heated to 82 °C overnight under continuous stirring. The solution was cooled, filtered and washed with water to remove remaining ammonia as a result of urea decomposition. Additionally, the catalysts were let to air-dry overnight followed by a higher temperature drying in an oven (100 °C) overnight. The final active form of the catalyst was achieved upon calcination under synthetic air (in case of Cu) or by reduction with pure hydrogen (in case of Ni, Co). The only parameters in the synthesis were the nature of the metal, the amount of it and the calcination or reduction temperature. The prepared catalysts were denoted according to the nominal metal loading excluding commercial HTC-500 catalyst as follows: HTC-500, Ni-19, Co-16, Cu-1.8 and Cu-4.5 (Tables 1 and 2).

Table 1

Preparation parameters of the investigated catalysts.

Catalyst	Preparation method	Optimized calcination or reduction temperature and atmosphere	Metal precursor
HTC-500	Used as received	No treatment	Unknown
Ni-19	Deposition precipitation with urea	Reduction at 500 °C pure hydrogen	NiNO_3
Co-16	Deposition precipitation with urea	Reduction at 500 °C pure hydrogen	CoNO_3
Cu-1.8	Deposition precipitation with urea	Calcination at 400 °C synthetic air	CuNO_3
Cu-4.5	Deposition precipitation with urea	Calcination at 400 °C synthetic air	CuNO_3

2.2. Evaluation of catalytic properties of synthesized catalysts for valorization of bio-ethanol to bio-butanol

The performance testing of different alumina supported metal catalysts was carried out in a laboratory-scale continuous stainless steel fixed bed reactor operated in liquid-phase (Fig. 1). In a typical experiment, 2 g of metal/alumina catalyst was placed in the reactor followed by heating to 240 °C and setting the reactor pressure to 70 bar (Ar). The reactor (length = 21 cm, diameter = 1.6 cm) was operated to upward flow and the packing of the reactor is depicted in Fig. 2. Thereafter, the HPLC-pump was turned on and the flow rate was adjusted to 0.1 ml/min. This facilitated the contact of the reactant (Etax Aa, 99.5% ethanol) with the catalyst bed and the product flow was cooled thereafter for sampling (room temperature). The sampling was carried out by a special valve capable of draining 1 ml sample. The treatment of the sample is described below. All catalysts were compared at the same steady-state temperature, pressure, flow rate, size fraction and mass. The LHSV (Liquid Hourly Space Velocity) value was 4.3(1/h) and ethanol specific velocity 8.3×10^{-6} (m/s).

2.3. Reaction product analysis

The reaction product analysis was carried out by dissolving 0.5 ml (from sampling valve) of product to 1 ml of tetrahydrofuran (Sigma Aldrich 33709) in gas chromatography (GC)-vial. The pre-treated samples were analyzed by means of GC-FID (Agilent Technologies 6890 N; column: DB-Petro 122-10A6, 100 m, i.d. 0.250 mm; flame ionization detector) calibrated with analytes.

2.4. Catalyst characterization

2.4.1. Structural analysis

Energy-filtered transmission electron microscopy (EFTEM, LEO 912 OMEGA, LaB₆ filament, 120 kV) was used to assess particle size and reveal the microstructure of the samples. The crystal phases of the support and catalyst particles are evaluated by X-ray powder diffraction (Bruker D8 Discover, Cu K α). From the broadening of the peaks in the patterns, Scherrer's equation was applied to estimate

Table 2

Measurement of the metal contents of catalysts using ICP-OES.

Catalyst	Nominal loadings (wt-%)	Measured (wt-%)	Rsd (%) ^a
Ni/Al ₂ O ₃ (Ni-19)	20	19	0.51
Co/Al ₂ O ₃ (Co-16)	20	16	2.16
Cu/Al ₂ O ₃ (Cu-1.8)	2	1.8	1.23
Cu/Al ₂ O ₃ (Cu-4.5)	5	4.5	0.53

^a Rsd (relative standard deviation) within three (3) replicates.

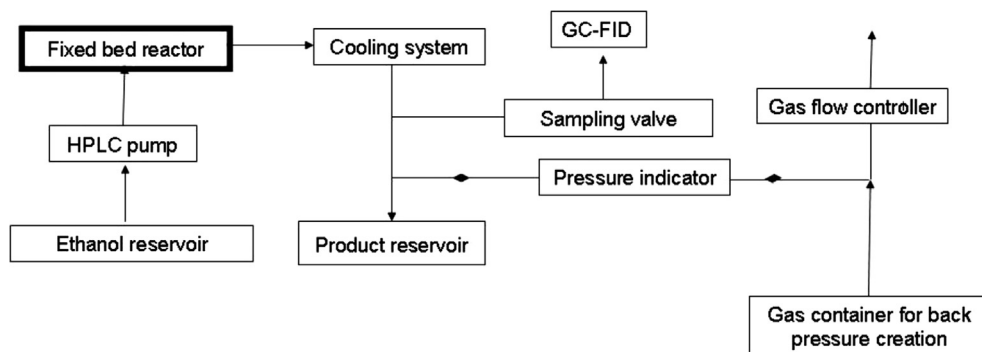


Fig. 1. Flowsheet of the experimental equipment.

the corresponding average crystallite size. The scale of the X-ray powder diffraction value 2θ was from 30 to 55°.

2.4.2. Fourier transform infrared spectroscopy (FT-IR)

FT-IR (ATI Mattson) was used to determine the Brønsted and Lewis acid sites of the prepared catalysts, using pyridine as a probe molecule. A self-compressed disc of alumina catalyst was placed into the FTIR cell followed by evacuation. The temperature was raised to 450 °C for 1 h and decreased to 100 °C for the measurement of the background spectrum. Pyridine was adsorbed to the catalyst for 30 min followed by subsequent desorptions at 250 °C, 350 °C and 450 °C, respectively. Moreover, sample spectra were recorded between each temperature ramp. The measurement was carried out under vacuum at 100 °C. Spectral bands at 1545 cm⁻¹

and 1450 cm⁻¹ were used to identify the corresponding Brønsted and Lewis acid sites. The quantitative characterization of both sites was evaluated with the aid of the constants of Emeis [36].

2.4.3. X-ray photoelectron spectroscopy (XPS)

XPS spectra were collected with a KRATOS Axis Ultra DLD electron spectrometer under monochromatic Al K α radiation (1486.6 eV). A pass energy of 160 eV with a step size of 1 eV was used for survey scans. For separate photoelectron lines, a pass energy of 20 eV was used with a step size of 0.1 eV. To compensate for the charging of the surface, a charge-neutralizing equipment was used, and the binding energy (BE) scale was referenced to the adventitious C 1s peak at 285.0 eV.

2.4.4. Inductively coupled plasma optical emission spectrometry (ICP-OES)

ICP-OES measurements were performed over self-prepared catalysts with Perkin Elmer Optima 5300 DV instrument. A sample of 55 mg was weighed into Teflon bomb followed by addition of acid mixture (3 ml H₂SO₄ 96% + 3 ml HNO₃ 65%). Acids were of suprapur grade. Thereafter, a microwave oven (Anton Paar, Multiwave 3000) was applied in the digestion (maximum 210 °C/1400 W). Dilution to 100 ml was performed followed by a further 10 times prior to the ICP analysis.

3. Results and discussion

3.1. Evaluation of catalytic properties of Co-, Ni- and Cu-modified alumina for valorization of bio-ethanol to bio-butanol

Based on extensive screening among different metal loadings as well as calcination and reduction temperatures, the best metals with corresponding loadings are presented here. The model catalyst (HTC-500, 20.7 wt-% Ni/Al₂O₃, Crossfield) presented before [33] was included for comparison with self-prepared alumina catalysts from prescreening results [33]. As a result of extensive screening, the three metals presented here were found to be the best metals when aiming at producing 1-butanol (Cu, Ni) or ethyl acetate (Co). The evaluation of the three most promising metals (Table 1) was carried out by varying the metal loading and calcination (Cu) or reduction temperature (Ni, Co). The optimized copper loadings used were at 1.8- and 4.5 wt-%, respectively (marked as Cu-1.8 and Cu-4.5), cobalt at 16 wt-% (Co-16) and nickel at 19 wt-% (Ni-19). The targeted and measured metal amounts are reported in Table 2.

Based on our observations, the optimal calcination temperature for copper catalysts was in the range of 400 °C, whereas for cobalt and nickel, the suitable reduction temperature was 500 °C. Interestingly, a shift to 100 °C higher or lower calcination temperature

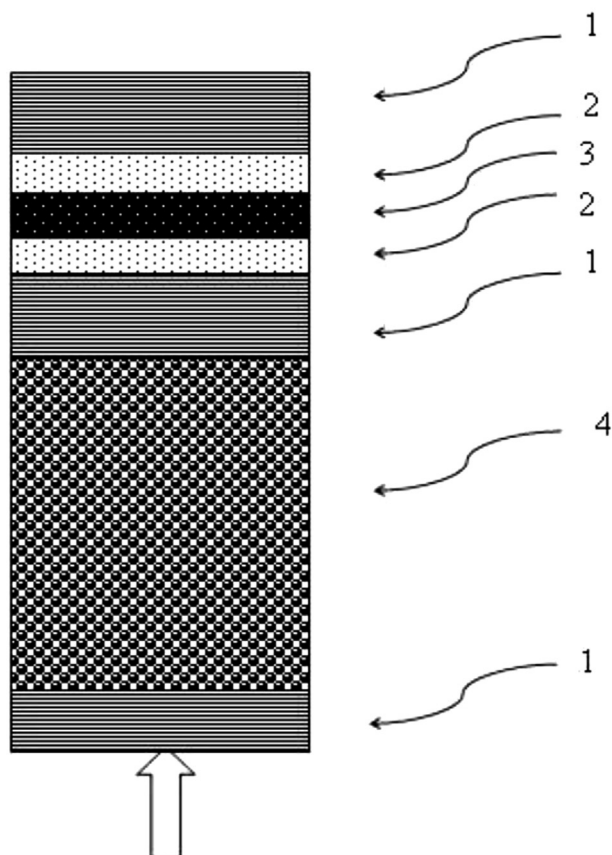


Fig. 2. Reactor packing; 1 = quartz wool, 2 = quartz sand, 3 = catalyst, 4 = silica beads.

resulted at an almost totally inactive copper catalyst. It is evident that during the heating, specific changes in the crystallite structure of the catalyst occur and, at a certain temperature, the active metal starts to interact with the alumina support so that the crystallite structure needed for reaction is formed. Additionally, cobalt and nickel catalysts were activated upon reduction at 500 °C (higher temperatures were not applied due to safety issues). The detailed explanation will be discussed later (Section 3.2). The conversion (Fig. 3, Equation (1)) and selectivity profiles (Fig. 4, Equation (2)) of the catalysts investigated under continuous operations are depicted in Figs. 3 and 4, respectively.

$$\text{Conversion} = \frac{c(\text{EtOH})_0 - c(\text{EtOH})_x}{c(\text{EtOH})_0} \times 100 \quad (1)$$

where

$c(\text{EtOH})_0$ = initial ethanol concentration (mmol/l) and
 $c(\text{EtOH})_x$ = measured ethanol concentration in the stream (mmol/l).

It can be observed that in the case of the commercial nickel catalyst (HTC-500), the conversion levels were in the range of 15% although it dropped over time to 10% due to catalyst deactivation. On the contrary, in case of Ni-19, stable level of 25% conversion was reached, whereas the cobalt catalyst (Co-16) gave rise to almost 30% conversion level. In addition, for copper (Cu-1.8, Cu-4.5) catalysts, the conversion level remained in the range of 12–15% regardless of the metal content. It may be that the mixed phase alumina support is capable of creating only a limited amount of catalytically active surface sites for ethanol condensation.

$$\text{Selectivity} = \frac{c(\text{analyte})_{x\nu}}{c(\text{EtOH})_0 - c(\text{EtOH})_x} \times 100 \quad (2)$$

where

ν = stoichiometric factor, $c(\text{analyte})$ = measured concentration of analyte (mmol/l), $c(\text{EtOH})_0$ = initial ethanol concentration (mmol/l) and $c(\text{EtOH})_x$ = measured ethanol concentration in the product steam (mmol/l).

The selectivity profiles obtained provide interesting information related to the characteristic metal properties. Both nickel catalysts

had similar selectivity profiles towards 1-butanol (60–65%). However, the tailor-made catalyst promoted an increase in gaseous products (H_2 , CO , CO_2 , CH_4) due to ability of metallic nickel on the catalyst surface to break carbon–carbon bond, albeit the conversion obtained was higher than in the case of the commercial catalyst. The formation of 1-hexanol can be attributed to the Guerbet reaction as reported earlier [33]. In addition, acetaldehyde is known to be a prerequisite for the Guerbet reaction. Overall, the nickel-based catalysts did not tend to form any ethyl acetate. The observed, apparent instability in the selectivity profiles is related to draining of the pressurized product reservoir, which upon periodical draining led to a temporary pressure drop. It seems that a drop in the pressure is followed by an increase in gaseous products by a simultaneous decrease in the selectivity towards liquid products. Unfortunately, the qualitative and quantitative determination of gaseous products is impossible at the current reactor system. However, our previous research [33] demonstrated the formation of mainly hydrogen and methane over HTC-500 catalyst.

Almost comparable selectivities towards 1-butanol was obtained over copper as nickel catalysts (Cu-1.8 = 65%, Cu-4.5 = 55%). The exception observed is related to the formation of ethyl acetate, which becomes prominent as the loading of Cu was increased. The formation of ethyl acetate is discussed in detail in the literature [27–32]. It is plausible that an increase in the copper loading results in a different surface structure in the presence of the alumina support. Moreover, copper may provide coordinatively unsaturated cationic sites with higher loadings, thus, leading to ethyl acetate formation. These results clearly indicate that minor differences in the catalyst microstructure steer the reaction pathway towards 1-butanol or ethyl acetate. It is also important to observe that the amounts of gaseous products in the presence of copper catalysts is minor, probably due to the ability of copper to preserve the carbon–carbon bonds. The most interesting result is represented by cobalt (Co-16) producing almost solely ethyl acetate with the selectivity of 85%.

The only previous work reporting γ -alumina [17] describes ethanol conversion levels of 19% and selectivity of 64% towards 1-butanol taking advantage of a catalyst with 8 wt-% nickel loading is in good correlation with our results. Also in this study, similar tendency of cobalt to form ethyl acetate was observed although in much lower amounts. It is evident that the crystallite structure explains at least partially the differences in γ - and mixed phase alumina. The complexity of alumina is well described in literature [37]. Additionally, the literature covering hydroxyapatite type structures [10–16,18–21] indicates that selectivities approaching 80% (1-butanol) are achievable. In addition, synthesis over low valence MgO has been presented by Coville et al. [22] yielding 1-butanol up to 20% and Davis et al. [23] with the selectivity of 34% towards 1-butanol.

3.2. Catalyst characterization results

The evaluation of the physico-chemical catalyst tested was carried out by several characterization techniques. XRD and TEM micrographs of the Ni-19 catalyst are presented in Fig. 5 which provides information about the structure and particle size. The presence of NiO was confirmed whereas the remaining metallic nickel is the obvious reason for the formation of gaseous products. However, upon comparison of the XRD of the commercial HTC-500 catalyst to our measurements performed earlier [33], some deviations can be noticed. HTC-500 catalyst indicates sharper diffractions, which can be explained by a different alumina support.

Further investigations were carried out over the cobalt (Co-16) catalyst (Fig. 6) to reveal its totally different product profile (ethyl acetate formation). XRD results show that the cobalt catalyst

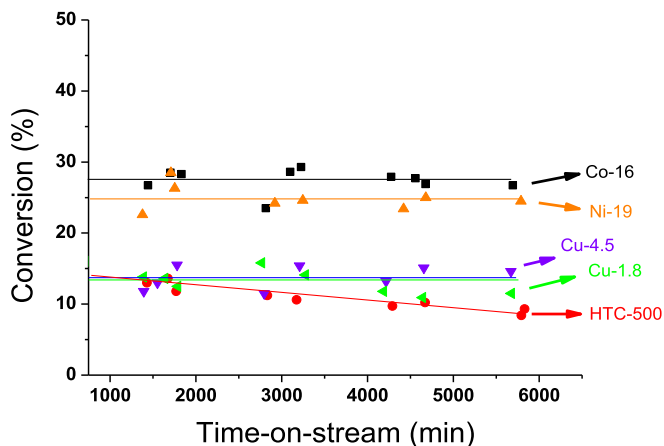


Fig. 3. Long-term conversion profiles of bio-ethanol over commercial HTC-500, Co-16, Ni-19, Cu-4.5 and Cu-1.8 catalysts. Points correspond sampling and the straight lines are averages.

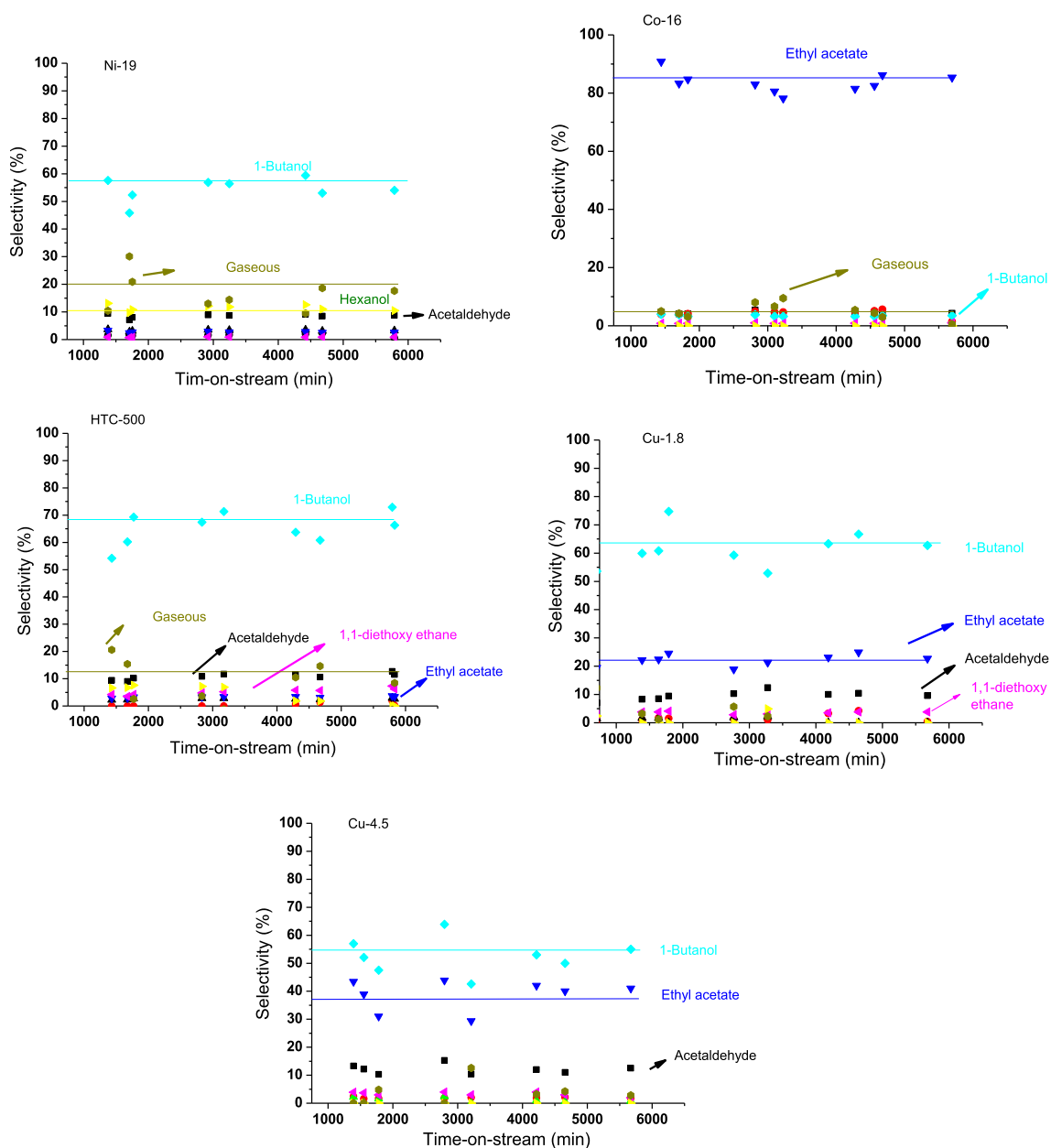


Fig. 4. Selectivity to main products 1-butanol, ethyl acetate and acetaldehyde over Ni-19, HTC-500, Co-16, Cu-1.8 and Cu-4.5 catalysts.

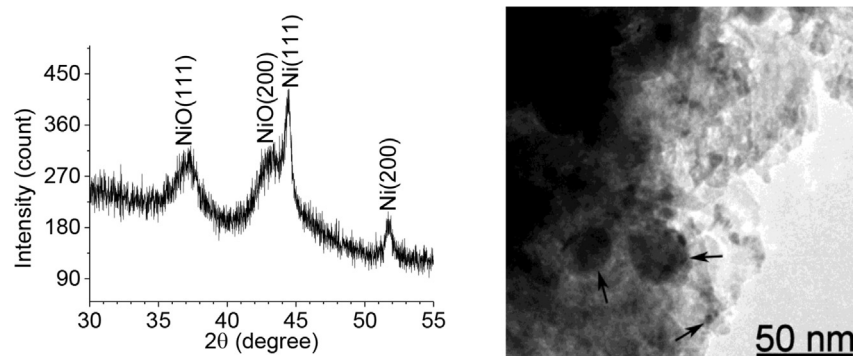


Fig. 5. X-ray diffraction patterns and transmission electron micrograph for Ni-19 catalyst (Ni 13.7 ± 0.2 nm; NiO 5.2 ± 0.7 nm).

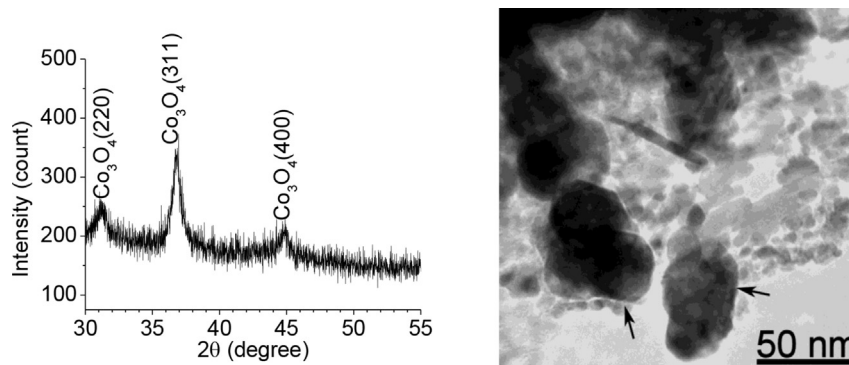


Fig. 6. Characterization of Co-16 (Co_3O_4 10.0 ± 2.6 nm) using X-ray powder diffraction and Transmission electron microscopy.

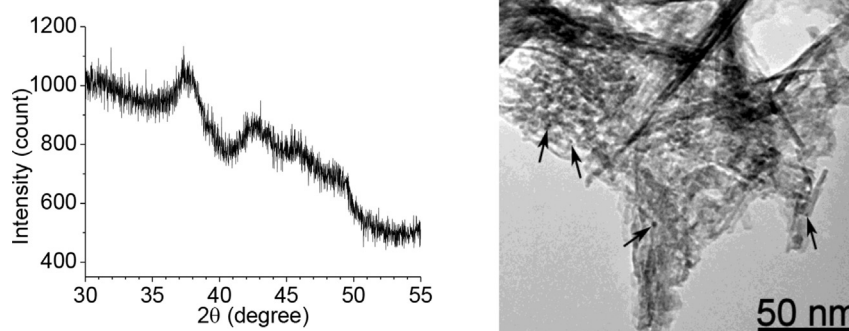


Fig. 7. Characterization of Cu-1.8 catalyst using X-ray powder diffraction (no reflections) and Transmission electron microscopy.

constitutes of a Co_3O_4 spinel [32] crystallite structure and, additionally, the peak profile is sharper in comparison to the Ni-19 catalyst indicating larger crystals in the specimen. Consequently, there seems to be different kind of interactions between the metal and the support that is the underlying reason to the different product composition which will be presented later.

Further evidence is provided by copper catalysts. For the low wt-% Cu-1.8 (Fig. 7) catalyst, only diffuse X-ray scattering was observed – if any – whereas the high wt-% Cu-4.5 (Fig. 8) gives rise to relatively sharp CuO-peaks. As illustrated in Fig. 4, higher loading of copper results in an increased ethyl acetate selectivity and reduced 1-butanol formation.

XPS measurements of different catalysts were performed to reveal the chemical states (or speciation) of metals and to correlate them with the catalytic results (Fig. 9).

Nickel catalysts depict rather similar spectra with the only exception being the double peak on the home-made Ni-19 catalyst. Analysis of the binding energy values (Table 3) and a comparison with existing literature data [28–43] results in the interesting observations. The peak at 856.3 eV corresponds to a nickel aluminate structure with octahedrally coordinated Ni^{2+} cations, these being the principal reason for the XRD peak broadening. By comparing the XPS signal (856.3 eV) of both nickel catalysts, it can be proposed that the laboratory-prepared catalyst provides more octahedrally coordinated Ni^{2+} aluminate structures responsible for the higher ethanol conversion. The observation is based on higher intensity of the laboratory-prepared nickel catalyst at 856.3 eV corresponding octahedral nickel aluminate coordination. Additionally, based on the literature [28–43], nickel is known to prefer tetrahedral coordination in lower nickel content alumina catalysts. Surprisingly,

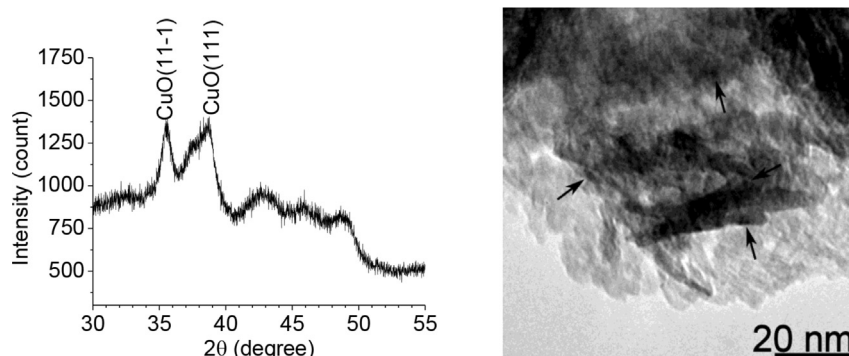


Fig. 8. Characterization of Cu-4.5 (Cu 8.5 ± 0.7 nm) using X-ray powder diffraction and Transmission electron microscopy.

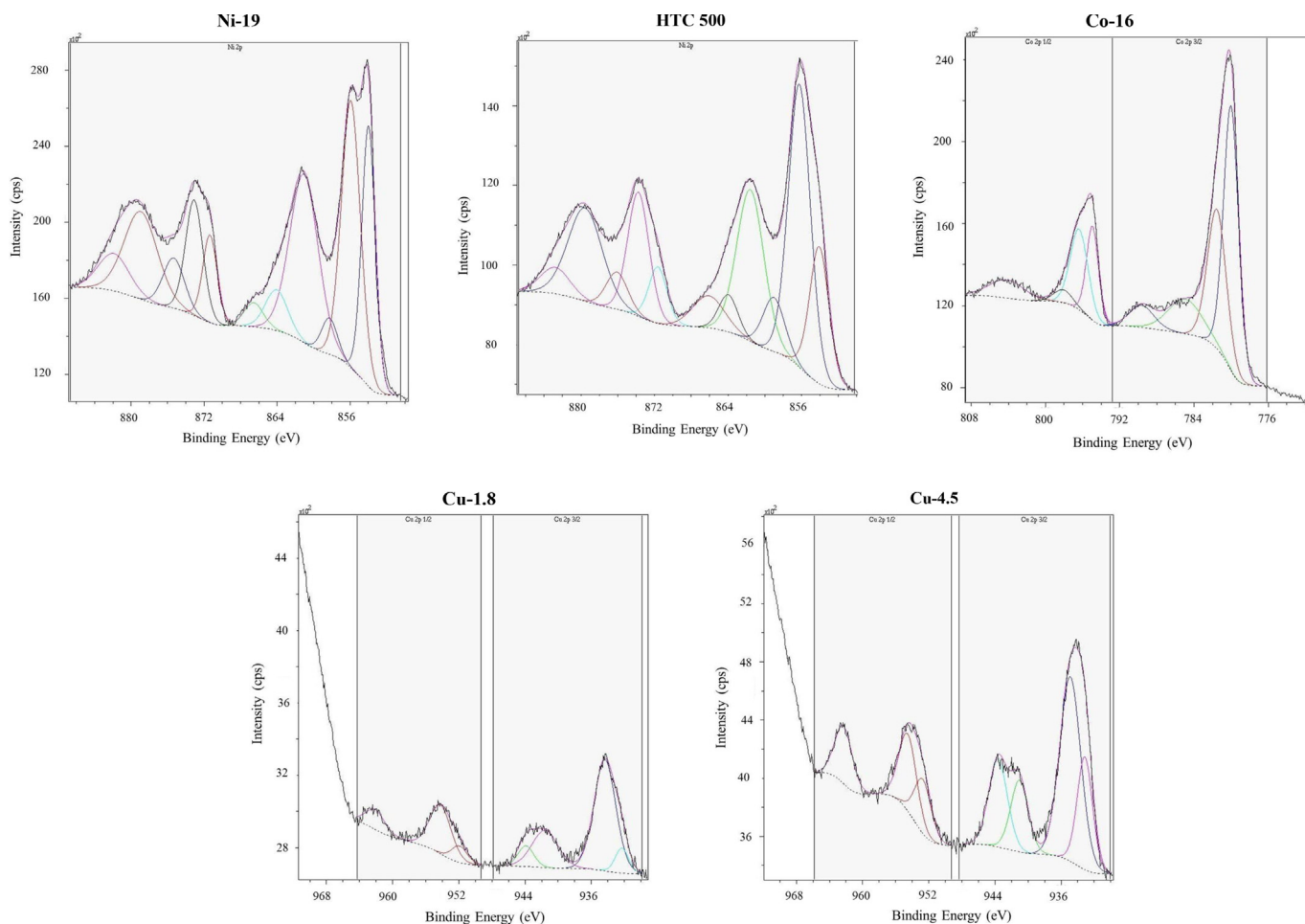


Fig. 9. XPS spectra of Ni-19, HTC-500, Co-16, Cu-1.8 and Cu-4.5 supported on alumina.

octahedral sites are preferred upon higher metal loadings, which confirms our observations (low loading results in no 1-butanol formation). It is also known that tetrahedrally coordinated nickel ions strongly interact with the alumina support, whereas octahedral coordination results in weak metal-support interaction.

By studying the information on cobalt supported alumina catalysts [44–49], an interesting observation can be made. Let us recall, the difference between the selectivity profiles of nickel and cobalt towards 1-butanol (Ni) and ethyl acetate (Co) (Fig. 4). The Co_3O_4 spinel structure is known to contain tetrahedrally coordinated Co^{2+} and octahedrally coordinated Co^{3+} . By connecting the observation to our results obtained with Co-16, it can be suggested that tetrahedral M^{2+} sites are responsible for the formation of ethyl acetate, whereas octahedral M^{2+} sites lead to 1-butanol.

The catalyst characterization results from copper catalysts (Cu-1.8, Cu-4.5) reveal useful information supporting our claim of the octahedral site preference for 1-butanol synthesis. At low loadings,

copper forms aluminate spinel, whereas a higher Cu-loading leads to an emergence of CuO sites, in combination with the spinel structure (XRD, XPS). However, it is suggested in literature [50–56] that an increase in the copper loading is followed by the formation of isolated Cu surface structures (in this case CuO). Moreover, the spinel structure is known to disappear upon an increase in the copper loading. The issue still remains what might be the explanation for the observed behavior between nickel and cobalt being active at high loadings in comparison to the low-loading activity of the copper. It is very probable that Cu^{2+} ion is affected by Jahn-Teller distortion in the spinel structure. More interestingly, the distortion is followed by an increase in the octahedrally coordinated copper cations, in line with our observations that this is the principal reason for the observed behavior changes upon various metal loadings. Thus, there seems to be a good correlation between the presence of octahedral M^{2+} sites

Table 3
Binding energy (E_b) of supported metals $2p_{3/2}$ peaks with corresponding structure.

Catalyst	Binding energy (E_b), eV	Corresponding structure
Ni-19	854.0/856.0	$\text{NiO}/\text{Ni}^{2+}$ octahedral aluminate
HTC-500	854.1/856.3	$\text{NiO}/\text{Ni}^{2+}$ octahedral aluminate
Co-16	780.0/781.5	$\text{Co}^{2+}/\text{Co}^{3+}$ in Co_3O_4 spinel
Cu-1.8	932.3/934.4	$\text{CuO}/\text{CuAl}_2\text{O}_4$ spinel
Cu-4.5	933.2/934.9	$\text{CuO}/\text{CuAl}_2\text{O}_4$ spinel

Table 4
Determination of Brønsted and Lewis acid sites with catalysts using FTIR-pyridine.

Catalyst	BAS (250 °C) (mmol/g _{cat})	BAS (350 °C) (mmol/g _{cat})	LAS (250 °C) (mmol/g _{cat})	LAS (350 °C) (mmol/g _{cat})
Ni-19	13	4	52	11
HTC-500	98	9	333	27
Co-16	0	14	73	3
Cu-1.8	11	5	52	88
Cu-4.5	16	9	87	138



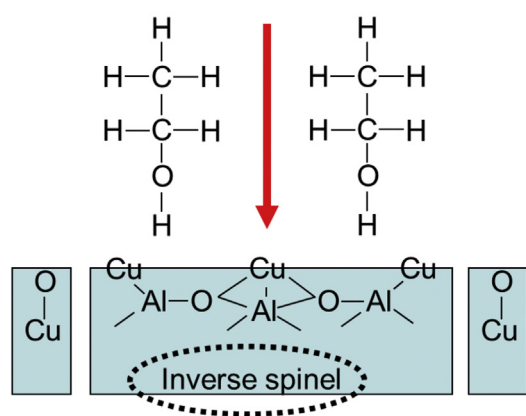
Fig. 10. Images of Cu-1.8 (left), Cu-4.5 (centre) and Cu 11 wt-% (right).

and the 1-butanol selectivity. It is also understood that mixed phase alumina [37] support contains more octahedrally coordinated aluminum ions than the traditional γ -alumina. Therefore, the presence of these could possibly facilitate the formation of surface aluminates. Nevertheless, some steric factors might affect the condensation between octahedral and tetrahedral sites since tetrahedral sites are known to be located a little deeper in the surface crystallite structure. Additionally, the observed catalytic behavior can also be explained by means of a site-preference energy concept which states that nickel and copper ions have a higher natural tendency to form octahedral aluminate compounds than cobalt [57].

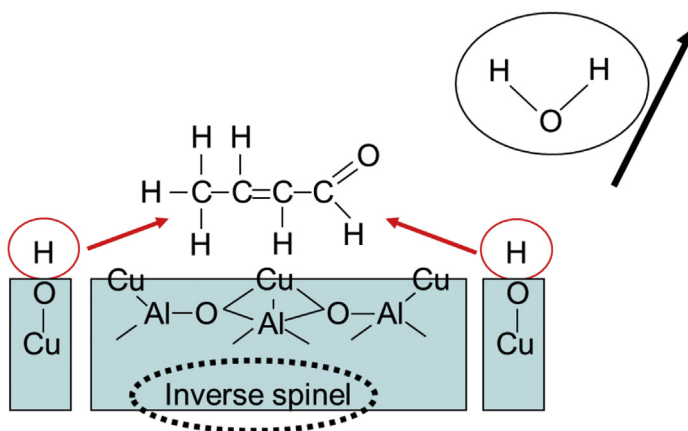
More interesting results were obtained from acidity measurements (Table 4). The amounts of weak Brønsted and Lewis acidic

sites were notably higher in the commercial HTC-500 catalyst than in the other ones. The observation confirms our view on a lower amount of octahedral surface aluminate sites present in the HTC-500 catalyst in comparison to Ni-19. The tetrahedral sites are known to have strong metal support interaction. It is also known that this is followed by an increase in acidity of the nickel cation in the aluminate structure, being in line with this work. Another important observation can be made e.g. between medium strength Lewis acid sites among the copper catalysts investigated. The increased acidity, resulting from the higher copper loading, is related to the disappearance of the aluminate structure and, thus, to a declining amount of octahedral surface sites.

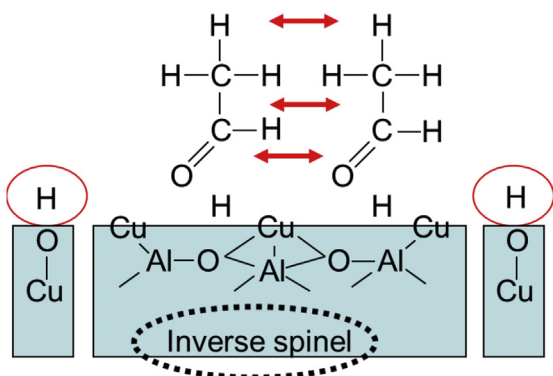
The effect of copper loading was studied experimentally (Fig. 10) by preparing a 11 wt-% Cu/alumina catalyst exactly with the same preparation method as Cu-1.8 and Cu-4.5 catalysts to investigate whether higher copper (CuO) content would have an effect on the investigated reaction. Result of the catalyst screening demonstrated that 11 wt-% copper catalyst was totally inactive supporting our view of metal aluminates as the catalytically active sites in the aldol condensation of ethanol to 1-butanol. Additionally, it can clearly be seen that the low loading Cu-1.8 (left) exhibits a green/turquoise color (in the web version) resulting from the aluminate phase, whereas higher loadings are in line with our results showing black Cu(II)O. Based on the results, there seems to be a clear correlation



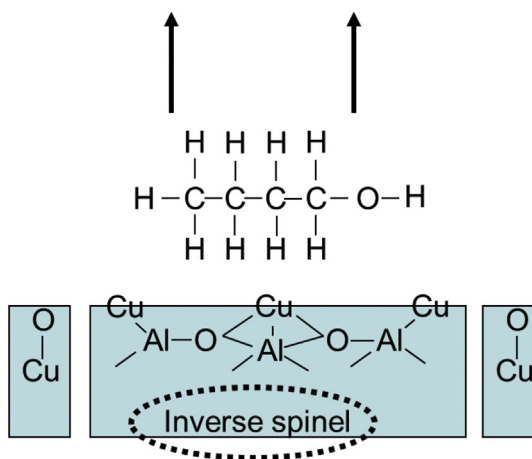
A. Adsorption of two ethanol molecules on the spinel surface.



C. Hydrogenation of formed crotonaldehyde intermediate to 1-butanol.



B. Oxidation of ethanol to acetaldehyde and aldol condensation between them. Accommodation of hydrogen on metal oxides.



D. Desorption of 1-butanol from the catalyst surface.

Fig. 11. A–D. Suggested reaction mechanism for aldol condensation of bio-ethanol to bio-butanol on alumina catalysts at 240 °C and 70 bar.

between octahedral surface aluminates and successful 1-butanol production.

3.3. Reaction mechanism and characteristic features

We have previously discussed mainly the issues related to the catalyst surface structure and acidity, but the ethanol molecule itself needs to be evaluated more precisely to gain a deeper understanding of the surface chemistry during the aldol condensation and the side reactions. Ethanol molecules naturally interact with the surface before undergoing aldol condensation between two surface-bound aldehyde molecules. The formed aldehydes origin from oxidized ethoxy groups, but it is important to remember that even minor modifications in the catalyst may result in totally different reaction products as discussed earlier. The ethanol adsorption chemistry [58–61] and reactivity among different metal oxides [62–64] is well covered in the literature, providing more support for the sensitiveness of the ethanol chemistry. For instance, ethanol may chemisorb as three different forms to the γ -alumina surface with only one structure leading to aldol condensation [58] and, thus, to 1-butanol. As a consequence, even minor modifications in the catalyst surface structure may have a remarkable effect on the product distribution.

The proposed need of adjacent acid-base sites in combination with strong base sites [65,66] can still be valid but we suggest that this needs to be addressed more precisely with crystallite chemistry as discussed above. The acetaldehyde condensation chemistry is investigated elsewhere [67], but it is important to keep in mind that the aldol condensation of acetaldehyde requires an appropriate acid-base strength [68]. The suitable catalytically active site including suitable acid-base strength is created on the spinel structure. More information related to acid-base bi-functionality [69,70] as well as base catalysis [71,72] is also available. Our suggestion for the possible reaction mechanism is schematically depicted in Fig. 11 A–D. However, the importance of the possible nearby CuO-site can be speculated.

In addition, the formation of ethyl needs to be more closely covered due to its simultaneous build up with 1-butanol. The selected open literature [27–32] provides a wide category of possible explanations on different catalysts and reaction conditions. For instance, Cu^+ [27] and pair of Cu^+/Cu^0 [31] have been proposed to promote the formation of ethyl acetate on the investigated reaction. However, we cannot fully combine the proposed effect with our results over the metals (Cu, Co, Ni) investigated. Our most promising hypothesis for the simultaneous manufacturing of ethyl acetate is based on an assumption that ethyl acetate is formed via a surface hemiacetal [28,29] compound. In this mechanism one ethanol molecule is oxidized to acetaldehyde which further reacts with another ethanol molecule by forming hemiacetal intermediate. Thereafter, hemiacetal is dehydrogenated to ethyl acetate (Fig. 12). Thus, the only difference between aldol condensation (crotonaldehyde) and hemiacetalisation reaction is based on whether two aldehydes (aldol condensation) or aldehyde and ethanol (hemiacetalisation) react together. The dependency between these four carbon atom intermediates is probably extremely sensitive to the crystallite structure of a catalyst and also to process conditions e.g. pressure.

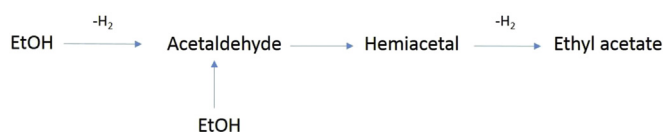


Fig. 12. Proposed reaction mechanism for the formation of ethyl acetate.

4. Conclusions

The real surface reaction mechanism leading to the transformation of ethanol to 1-butanol (at 240 °C and 70 bar) seems to be based on the aldol condensation of two aldehyde molecules as reported earlier. However, the proposed need of adjacent acid-base sites needs to be critically addressed. It is proposed that the catalytically active site for 1-butanol formation is based on the inverse spinel metal aluminate structure containing octahedrally coordinated metal cations which were present in Cu- and Ni-catalysts, while tetrahedrally coordinated cations (Cu, Co) result in the formation of ethyl acetate. The difference between these two crystallite surface structures can be explained by differences in the strength of the metal-support interaction, known to be strong in tetrahedra but weak in octahedra.

Forthcoming research will cover the ongoing work among the preparation of totally self-prepared metal oxide catalysts and evaluation of the most promising catalyst candidates in the continuous fixed bed reactor reported here for the synthesis of 1-butanol from bio-ethanol. Additionally, the effect of aqueous ethanol solutions will be studied both from fundamental and industrial viewpoint.

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